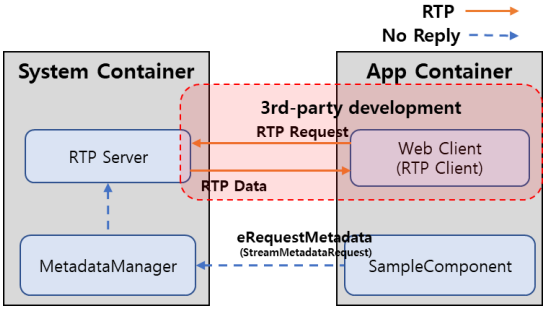
send_metadata

This sample application is designed to test the send-metadata-event.

Overview: Event Flow

The application can send the [Event metadata] and [Frame metadata] events to the camera. [Event metadata] is generated when an event occurs, which contains information about the topic, timestamp, source, data, etc. [Frame metadata] is generated when every frame of video data is generated, which contains information about the resolution, timestamp, object, etc.



- Possible Scenario:

  Sending an event and subsequently requesting and receiving metadata via the Real-time Transport Protocol (RTP) is a possible scenario.

Data Types for Event

- **Common Parameters**
  - StreamMetadata

Function	Return	Parameter	Description
set_frame_metadata	void	EventMetadataItem& & event_metadata	Set frame_metadata
frame_metadata	FrameMetadataItem&	-	Returns frame_metadata
set_ptz_metadata	void	PtzMetadataItem& ptz_metadata	Set ptz_metadata.
ptz_metadata	PtzMetadataItem&	-	Returns ptz_metadata
set_event_metadata	void	vector& event_metadata	Set event_metadata
add_event_metadata	void	EventMetadataItem& & event_metadata	Add event_metadata
event_metadata	vector&	-	Returns event_metadata
NeedBroadcasting	void	bool broadcast	Set broadcast

Function	Return	Parameter	Description
SetSyncInfo	void	uint64_t profile_id, uint64_t session_id	Set Sync information.
SetSessionID	void	uint64_t session_id	Set session_id
GetSessionID	uint64_t	-	Returns session_id
SetProfileID	void	uint64_t profile_id	Set profile_id
GetProfileID	uint64_t	-	Returns profile_id
HasFrame	bool	-	Returns whether frame_metadata is present or not
HasPtz	bool	-	Returns whether ptz_metadata is present or not
HasEvent	bool	-	Returns whether event_metadata is present or not
IsBroadcastable	bool	-	Check whether it is Broadcastable
IsSyncable	bool	-	Check whether it is Syncable

- **Event Metadata Parameters**
  - EventMetadataItem

Function	Return	Parameter	Description
set_timestamp	void	uint64_t timestamp	Set TimeStamp
timestamp	uint64_t	-	Returns TimeStamp
set_topic	void	string&& topic	Set Tocic
topic	string&	-	Returns topic
set_property	void	string&& property	Set property
property	string&	-	Returns property
add_source	void	SimpleItem&& source	Add source
source	vector&	-	Returns source
add_key	void	SimpleItem&& key	Add key
key	vector&	-	Returns key
add_data	void	SimpleItem&& data	Add data
data	vector&	-	Returns data

Function	Return	Parameter	Description
add_element	void	ElementItem&& element	Add element
element	vector&	-	Returns element

- SimpleItem

Variable	Type	Description
name	string	Simple item’s key
value	string	Simple item’s Value

- ElementItem

Variable	Type	Description
element_name	string	Element’s name
descriptor_name	string	Descriptor’s name
element_data	string	Element data
candidate	vector	candidate’s struct

- Candidate

Variable	Type	Description
type_name	string	Candidate’s Name
likelihood	float	Candidate’s likelihood

- **Frame Metadata Parameters**
  - FrameMetadataItem

Function	Return	Parameter	Description
set_timestamp	void	uint64_t timestamp	Set TimeStamp
timestamp	uint64_t	-	Returns TimeStamp
set_resolution	void	int width, int height	Set width and height
width	int	-	Returns width
height	int	-	Returns height
set_media_source_token	void	string& token	Set media_source_token
media_source_token	string&	-	Returns media_source_token

Function	Return	Parameter	Description
set_video_source_token	void	string& token	Set video_source_token
video_source_token	string&	-	Returns video_source_token
set_geolocation	void	bool enable, float lon, float elevation	Set geolocation
geolocation	metadata::common::Geolocation&	-	Returns geolocation
add_object	void	metadata::common::Object&& object	Add object
object	vectormetadata::common::Object&	-	Returns object

- Object

Variable	Type	Description
rule_id	uint32_t	rule id info
object_id	uint32_t	object id info
type	MetadataType	type info
category	ObjectCategory	Category info(0 ~ 14)
rect	Rect	Rect info
likelihood	float	likelihood info
parent_id	uint32_t	parent id info
attr_flag	uint32_t	attribute flag info
attr_num	uint32_t	attribute number info
attr_class	uint64_t	attribute class info
timestamp	uint64_t	TimeStamp info
crop_img_info	Rect	crop image info
proximate_obj_id	uint32_t	proximate object id info
proximate_distance	float	proximate distance info
image_ref	string	image reference info

- Rect

Variable	Type	Description
sx	int32_t	Starting point of x
sy	int32_t	Starting point of y

Variable	Type	Description
ex	int32_t	Ending point of x
ey	int32_t	Ending point of y

Setting EventMetadata

- Create EventMetadataItem.

```
auto event_metadata = EventMetadataItem();
```

- Set the topic and timestamp of EventMetadataItem.

```
event_metadata.set_topic("tns1:VideoSource/tnssamsung:" + <App Name>);
event_metadata.set_timestamp(timestamp);
```

- Add the source and data of EventMetadataItem.

```
event_metadata.add_source({"VideoSourceToken", "VideoSourceToken-0"});
event_metadata.add_data({"State", "true"});
```

- You can add ElementItem to EventMetadataItem.To add ElementItem, set it as follows.
  - Set element_name and descriptor_name of ElementItem.

    ```
    EventMetadataItem::ElementItem element_item;
    element_item.element_name = "sample_element_name";
    element_item.descriptor_name = "sample_descriptor_name";
    ```

  - Set type_name and likelihood of Candidate.

    ```
    EventMetadataItem::Candidate candidate;
    candidate.type_name = "sample_type_name";
    candidate.likelihood = 0.94;
    ```

  - Add Candidate to ElementItem.

    ```
    element_item.candidate.push_back(candidate);
    ```

  - Add ElementItem to EventMetadataItem.

```
event_metadata.add_element(std::move(element_item));
```

- Create StreamMetadata using Channel and Timestamp.

```
auto metadata = StreamMetadata(channel, timestamp);
```

- Add EventMetadataItem to StreamMetadata.

```
metadata.add_event_metadata(std::move(event_metadata));
```

- Set Broadcasting in StreamMetadata.

```
metadata.NeedBroadcasting(true);
```

Setting FrameMetadata

- Create a FrameMetadataItem.

```
auto frame_metadata = FrameMetadataItem();
```

- Set the resolution and timeStamp of FrameMetadataItem.

```
frame_metadata.set_resolution(3840, 2160);
frame_metadata.set_timestamp(timestamp);
```

- Set the data of the object.

```
metadata::common::Object send_object;
send_object.object_id = 2;
send_object.parent_id = 0;
send_object.timestamp = timestamp;
send_object.likelihood = (float)0.994629;
send_object.category = (metadata::common::ObjectCategory)0;
send_object.attr_num = 0;
```

- Set the Rect information of the object.

```
send_object.rect.sx = 85; // Starting point of x.
send_object.rect.ex = 120; // Ending point of x.
send_object.rect.sy = 68;   // Starting point of  y
send_object.rect.ey = 137;  // Ending point of y
```

- Create StreamMetadata using Channel, Timestamp, and MetadataType.

```
auto metadata = StreamMetadata(GetChannel(), timestamp,
Metadata::MetadataType::kAI);
```

- Set FrameMetadataItem to StreamMetadata.

```
metadata.set_frame_metadata(std::move(frame_metadata));
```

Sending StreamMetadata

- Set StreamMetadata to the builder.

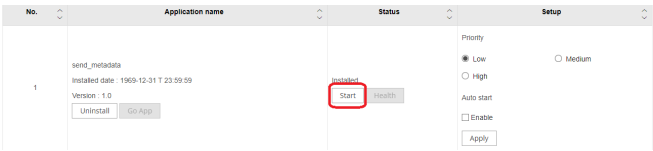
```
auto builder = IPMetadataManager::StreamMetadataRequest::builder();
builder.set_stream_metadata(std::forward<class
StreamMetadata>(metadata));
```

- Set builder to StreamMetadataRequest.
- Include StreamMetadataRequest in the event and send it to "MetadataManager".

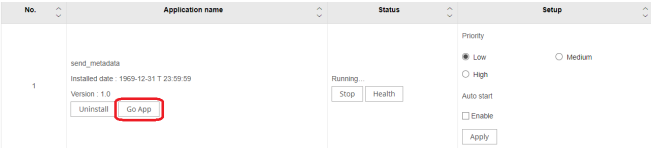
```
auto metadata_request =
reinterpret_cast<IPMetadataManager::StreamMetadataRequest*>(builder.bu
ild());
SendNoReplyEvent("MetadataManager",
static_cast<int32_t>(IMetadataManager::EEventType::eRequestMetadata),
0, metadata_request);
```

Sending Metadata

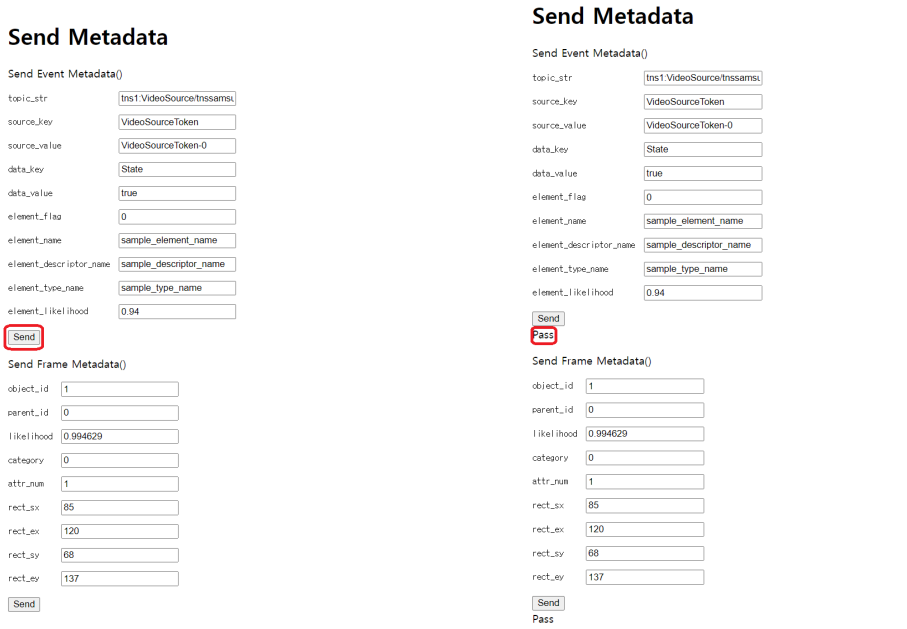
- Click [Start] on the web page.



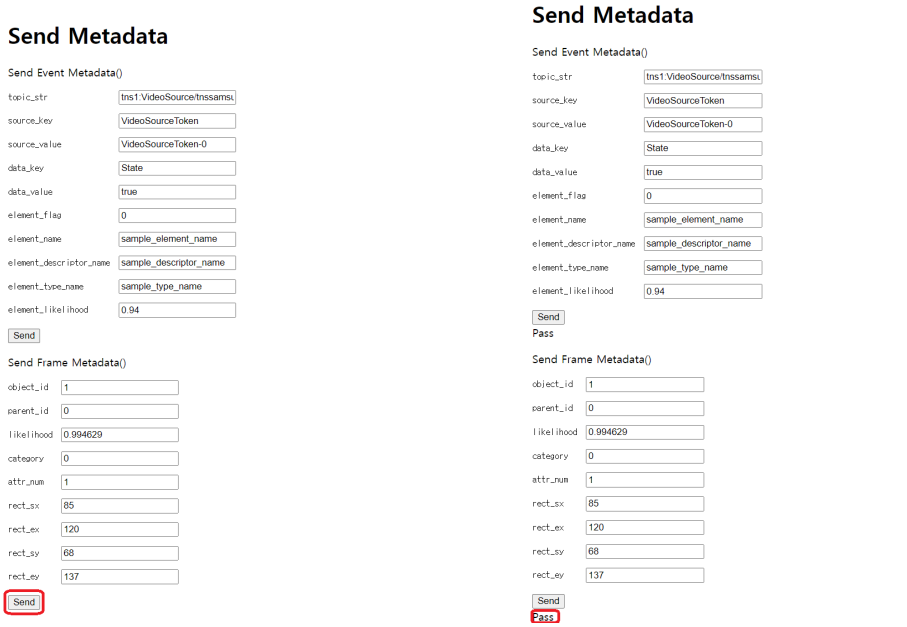
- Once the application is running, click [Go App].



- To send Event Metadata, click [Send] below [Send Event Metadata()].



- To send Frame Metadata, click [Send] below [Send Frame Metadata()].



xml from metadata

When an RTP Session is established and you click [Send], Event metadata as shown below is sent to RTP Meta Track.

- Event Meta

```
<?xml version="1.0" encoding="UTF-8"?><tt:MetadataStream
```

```
xmlns:tt="http://www.onvif.org/ver10/schema"
xmlns:ttr="https://www.onvif.org/ver20/analytics/radiometry"
xmlns:wsnt="http://docs.oasis-open.org/wsn/b-2"
xmlns:tns1="http://www.onvif.org/ver10/topics"
xmlns:tnssamsung="http://www.samsungcctv.com/2011/event/topics"><tt:Ev
ent><wsnt:NotificationMessage><wsnt:Topic
Dialect="http://www.onvif.org/ver10/tev/topicExpression/ConcreteSet">t
ns1:VideoSource/tnssamsung:send_metadata</wsnt:Topic><wsnt:Message><tt
:Message UtcTime="2000-01-25T02:24:20.613Z"><tt:Source><tt:SimpleItem
Name="VideoSourceToken" Value="VideoSourceToken-
0"/></tt:Source><tt:Key></tt:Key><tt:Data><tt:SimpleItem Name="State"
Value="true"/></tt:Data></tt:Message></wsnt:Message></wsnt:Notificatio
nMessage></tt:Event></tt:MetadataStream>
```

When an RTP Session is established and you click [Send], Frame metadata as shown below is sent to RTP Meta Track.

- Frame Meta

```
<?xml version="1.0" encoding="UTF-8"?><tt:MetadataStream
xmlns:tt="http://www.onvif.org/ver10/schema"
xmlns:ttr="https://www.onvif.org/ver20/analytics/radiometry"
xmlns:wsnt="http://docs.oasis-open.org/wsn/b-2"
xmlns:tns1="http://www.onvif.org/ver10/topics"
xmlns:tnssamsung="http://www.samsungcctv.com/2011/event/topics"
xmlns:fc="http://www.onvif.org/ver20/analytics/humanface"
xmlns:bd="http://www.onvif.org/ver20/analytics/humanbody"><tt:VideoAna
lytics><tt:Frame UtcTime="2000-01-
25T03:37:54.473Z"><tt:Transformation><tt:Translate x="-1.0"
y="1.0"/><tt:Scale x="0.000521" y="-
0.000926"/></tt:Transformation><tt:Object
ObjectId="2"><tt:Appearance><tt:Shape><tt:BoundingBox left="1685.0"
top="68.0" right="1920.0" bottom="374.0"/><tt:CenterOfGravity
x="1802.5"
y="221.0"/></tt:Shape><tt:Class><tt:ClassCandidate><tt:Type>Other</tt:
Type><tt:Likelihood>0.99</tt:Likelihood></tt:ClassCandidate><tt:Type
Likelihood="0.99">Unknown</tt:Type></tt:Class></tt:Appearance></tt:Obj
ect></tt:Frame></tt:VideoAnalytics></tt:MetadataStream>
```

Building Application

1. Build application.

```
$ APP_NAME=send_metadata SDK_VER=24.06.14(your SDK version) SOC=[cv5,
jetson] docker compose up
$ docker compose down --remove-orphans
```

2. Check the build results in current directory. If successful, you will be able to find the cap file.